

ESSE REPORT

ENVIRONMENTAL PROBLEM: ENVIRONMENTAL POLLUTION
REPORTING

PROPOSED ENGINEERING/IT SOLUTION: SMART
POLLUTION ALERT & NOTIFICATION SYSTEM

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ABSTRACT

Environmental pollution poses significant sustainability challenges to human health, the environment, and natural resources. Cases of air pollution, water pollution, waste disposal, soil pollution, and noise pollution are common in residential and urban areas. However, most of these occurrences are only visible to the human eye but rarely are reported or monitored by the appropriate authorities and local inhabitants. Current complaint systems only deal with complaints and do not offer any form of immediate location-based alerts or updates to the affected systems.

Therefore, this paper proposes a Smart Pollution Alert and Location-Based Environmental Notification System. The idea is to come up with an IT system that offers immediate information dissemination on environmental pollution incidents reported. The proposed system will have various features, including collecting the required information about the incident, classifying the type and severity of the pollution cases, and offering relevant information to the affected locations. Additionally, the system will be able to store the details retrieved about the pollution event in a database and used to generate statistics and reports.

It is envisioned that the pollution alert system will be developed as a responsive website that can be accessed through desktop and mobile devices. It will also be possible to monitor and include all the relevant information about the pollution incident detected. The system will open new dimensions in environmental monitoring, management, and pollution reporting while promoting sustainable development and supporting selected SDGs and environmental management principles. Furthermore, future implementations can include the use of GIS and IoT to monitor and collect real-time data using sensors and data analytics.

INTRODUCTION

Environmental pollution is a serious threat to the ecosystem that consists of atmosphere, hydrosphere, lithosphere, and other life forms. The main causes of pollution are rapid urbanization, industrialization, traffic, construction activities, and inappropriate waste disposal.

The pollution incidents can be of various types, such as environmental pollution caused by greenhouse gases, water pollution caused by chemical wastes, land pollution due to the accumulation of non-biodegradable materials, and noise pollution due to transportation and industrial activities. Human health and the environment are significantly affected by these incidents.

The problem with the current complaint reporting system is that most of the people witness the pollution incidents, but they do not have an effective way to report the incidents. Most complaint reporting systems focus only on collecting complaints and not on providing the necessary information about the complaint.

Information technology can play a significant role in this scenario. A web-based application can be developed to collect, store, process, and share the required information about the pollution incidents. The web-based application can utilize location-based services to provide the relevant information to the users.

The above-mentioned idea explains the concept of the web-based pollution reporting system, which focuses on providing a platform for sharing the pollution incident reports and relevant details. The system helps to increase the awareness among the people about the adverse effects of pollution and promotes an environment-friendly society.

PROBLEM STATEMENT

Environmental pollution cases like illegal dumping, excessive smoke, water contamination, industrial emissions, and noisy disturbances may have adverse effects on the surrounding community or environment. Despite the fact that the citizens may identify these problems, the current reporting and communication methods do not offer a direct way to inform the affected people in the vicinity promptly.

The traditional complaint mechanism focuses on complaint registration and collection. However, registering a complaint does not directly inform the citizens of an environmental pollution case in their area. Additionally, the information may be fragmented in different complaint management systems, thus, being unavailable to the ordinary user searching for pollution incidents in a specific region.

Since pollution incidents are diverse, the impact they create is also different depending on the pollution category and location. For example, a minor waste case may be less hazardous compared to a severe air pollution incident. Therefore, an efficient complaint management system should offer differentiated information based on pollution severity and category.

Therefore, there is a need for an IT-based solution to collect the details of a pollution incident, classify it into a particular category, define the affected area, and notify the users in this location about the problem. The Smart Pollution Alert and Location-Based Environmental Notification is a system designed to address this issue.

SUSTAINABILITY ANALYSIS

The offered system excellently meets the needs of environmental sustainability by improving environmental awareness, communication, and involvement in processes that address the pollution problem.

Sustainable Development Goals (SDGs)

SDG 3 – Good Health and Well-Being: The system can alert people about environmental hazards, thus preventing health issues.

SDG 6 – Clean Water and Sanitation: The system can collect information about water pollution incidents and provide this data for communication purposes.

SDG 11 – Sustainable Cities and Communities: The system can raise awareness about community environmental issues.

SDG 12 – Responsible Consumption and Production: The system can promote sustainability by reminding people about the need for responsible consumption and production of waste.

SDG 13 – Climate Action: The system can increase people's environmental awareness and involvement in addressing climate change issues.

SDG 15 – Life on Land: The system can collect reports about land pollution incidents and provide this information for communication purposes.

ISO 14001

The offered software can be integrated into an Environmental Management System (EMS) based on ISO 14001:2015. It would perfectly collect, store, analyze, and communicate the information about pollution incidents, as well as organize the necessary corrective actions.

LCA and EIA

Since the proposed solution is primarily software, it has a very limited Life Cycle Assessment (LCA). The electricity used to power the servers hosting the application, as well as the personal computers and other network equipment used to access it, comprise its main environmental impact. At the same time, no significant Environmental Impact Assessment (EIA) is needed because the system does not involve any large-scale construction or industrial activities.

PROPOSED ENGINEERING/IT SOLUTION

The proposed solution is a responsive web-based Smart Pollution Alert and Location-Based Environmental Notification System. It provides location-specific environmental alerts instead of only collecting pollution complaints.

The system follows this process:

Pollution Report → Verification → Classification → Severity Assessment → Location Identification → Alert Generation → User Notification

Users can report incidents by entering the pollution type, location, description, date, and supporting images. Administrators verify reports before publishing them. Incidents are classified as air, water, soil, noise, or waste pollution and assigned severity levels such as Low, Moderate, High, or Critical.

Key features include user registration, pollution reporting, category selection, location-based alerts, environmental notifications, alert history, pollution information, and an administrator dashboard. Administrators can verify reports, classify incidents, assign severity, publish alerts, and update incident status.

The database stores user, incident, location, alert, and notification information. The system can be developed using HTML, CSS, JavaScript, a suitable backend technology, and a database. Future enhancements may include GIS mapping, IoT sensors, real-time pollution monitoring, and automated data analysis.

FEASIBILITY ANALYSIS

The proposed system is technically, economically, operationally, and socially feasible.

Technical Feasibility:

The system can be developed using commonly available technologies such as HTML, CSS, JavaScript, backend technologies, and databases. Its responsive design allows it to work on computers and mobile devices.

Economic Feasibility:

The basic system does not require expensive physical infrastructure. Development can be carried out using freely available software tools and existing computer systems, keeping the initial cost low.

Operational Feasibility:

The system can be accessed through a web browser. Users can easily report incidents and view alerts, while administrators can manage reports through a dashboard.

Social Feasibility:

The system can encourage public participation and improve awareness of local environmental issues. However, verification of reports is important to reduce false or inaccurate information.

The system can initially be implemented on a small scale and later expanded to cover larger areas. Additional features such as GIS, IoT sensors, and real-time monitoring can also be integrated as required.

Overall, the proposed system is technically practical, economically affordable, operationally manageable, and socially useful.

EXPECTED ENVIRONMENTAL BENEFITS

The proposed system can provide several environmental and social benefits.

Improved Environmental Awareness:

Users receive information about pollution incidents in their surroundings and become more aware of local environmental problems.

Timely Communication:

Location-based alerts provide relevant information to people in affected areas.

Increased Public Participation:

Users can report pollution incidents and contribute to environmental monitoring.

Better Pollution Monitoring:

Stored records allow administrators to track incidents and monitor recurring problems.

Identification of Pollution Hotspots:

Repeated incidents in the same location can help identify areas requiring greater attention.

Support for Preventive Action:

Users can receive information about environmental risks and suitable precautions.

Reduced Paper Usage:

Digital reporting and notifications reduce dependence on paper-based complaint systems.

Better Decision-Making:

Collected data can help authorities understand recurring pollution problems and plan appropriate actions.

The system does not directly remove pollution. Instead, it acts as an information and communication platform that supports environmental monitoring, awareness, and timely response.

JUSTIFICATION OF THE PROPOSED SOLUTION

Effective environmental management requires proper monitoring, reporting, and timely communication. Conventional complaint portals mainly collect information from users, whereas the proposed system also communicates relevant pollution alerts to people in affected locations.

Location-based notifications make environmental information more relevant to users because they can receive alerts related to their surroundings. Severity classification also helps distinguish between different levels of pollution and identify incidents that may require greater attention.

The proposed system is suitable as an IT-based engineering solution because it combines web development, database management, location services, data processing, and notification features. It can be implemented as a responsive website without requiring expensive physical infrastructure.

The solution is also scalable. Future technologies such as GIS mapping, IoT sensors, real-time pollution monitoring, and data analytics can be integrated to improve its functionality.

Therefore, the proposed system provides a practical, user-oriented, and scalable approach to environmental pollution communication. It can improve public awareness and participation while supporting better documentation and monitoring of pollution incidents.

CONCLUSION

Environmental pollution requires effective monitoring, public awareness, communication, and timely action. Existing reporting mechanisms may collect pollution complaints but may not provide timely location-specific information to nearby communities.

The proposed Smart Pollution Alert and Location-Based Environmental Notification System addresses this problem through a responsive web-based platform. It allows pollution incidents to be reported, verified, classified according to type and severity, and communicated to users in affected areas.

The system can improve environmental awareness, public participation, and pollution monitoring. It also supports several Sustainable Development Goals and can contribute to environmental management activities related to ISO 14001.

The proposed solution is technically and economically feasible because it can be developed using commonly available web technologies without requiring expensive physical infrastructure.

Future improvements such as GIS mapping, IoT sensors, real-time monitoring, and data analytics can make the system more effective. Overall, the proposed solution provides a practical and scalable IT-based approach for improving environmental communication and supporting sustainable communities.

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